
Education

University of Utah

B.S. Computer Science

Salt Lake City, UT

August 2020 – May 2025

Relevant Coursework: Computer Systems, Machine Learning, Computer Graphics, Algorithms, Software Practice I & II, Database Systems, Computer Networks, Foundations of Data Analysis, Models of Computation, and Linear Algebra.

Skills

Languages: Python, TypeScript, C++, SQL, Java

Software: Pandas, NumPy, React, Node.js, AWS, Azure, Docker, Next.js, .NET, REST, Django, MongoDB, Linux

Experience

L3Harris Technologies

Associate Software Engineer

Dallas, TX

June 2025 – Present

- Reduced deployment friction across three production services by building containerized microservices in Python and C++ using Podman on AWS, standardizing CI/CD pipelines with static analysis and integration testing via GitHub Actions.
- Improved data throughput for high-frequency sensor ingestion pipelines by applying system-level profiling and low-level optimization on embedded Linux systems, meeting strict latency constraints in collaboration with hardware teams.
- Accelerated release confidence by automating validation pipelines with integrated testing and infrastructure-as-code tooling, enabling consistent multi-environment deployments across production.

Doxy.me

Software Engineer

Charleston, SC

August 2024 – May 2025

- Reduced manual clinical review overhead by building end-to-end ML data pipelines in Python and TypeScript that processed unstructured WebRTC video streams through PyTorch and Llama inference, surfacing structured outputs to physicians via SageMaker and Fargate.
- Improved diagnostic accuracy by implementing reinforcement learning feedback loops that closed the gap between live production signals and model behavior across deployed inference services.
- Enabled zero-downtime releases across multiple production environments by owning YAML-based CI/CD configuration and containerized AWS infrastructure, reducing deployment time and environment drift.

University of Utah

Undergraduate Research Assistant – FuTURES Lab

Salt Lake City, UT

May 2024 – Jun 2025

- Analyzed and optimized large-scale datasets, enhancing software configuration testing with gcov and CMake, increasing code coverage by 30% on real-world APIs (Libpng, PyTorch).
 - Expanded OSS-Fuzz testing coverage using compile-time options to improve the robustness of full-stack libraries.
-

Projects

University Course Planning/Review Platform (Class Best): Built a multi-service data platform enabling student schedule matching, course overlap detection, and behavioral insights. Developed ETL pipelines in Python to process thousands of course records, user preferences, and time-series activity logs. Built a responsive frontend in React + Tailwind with interactive dashboards, heatmaps, and calendar visualizations.

Telehealth Mental Health Diagnosis Tool (Doxy.me): Built an AI system that analyzes visual, speech, and tonal cues from telehealth sessions using PyTorch and Llama, delivering structured mental health insights to treating physicians via SageMaker and Fargate inference pipelines. Improved diagnostic accuracy using reinforcement learning and WebRTC video integration.

Configuration Fuzzer: Developed a configuration fuzzing tool for OSS-Fuzz, automating build generation to identify critical compile-time configurations and improve code coverage, uncovering previously untested code paths and increasing bug detection effectiveness.